Department of Information Technology

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch : III Semester-B. Tech Information Technology
Course Code & Title : CS3391 Object Oriented Programming
Name of the Faculty member (s) : S. Sakkaravarthi, AP/IT

Innovative Practice Description

- **Unit / Topic : Unit 5 / JavaFX Layouts-GridPane**
- **Course Outcome :** CO5
- **Topic Learning Outcome:** TLO17
- **Activity Chosen : Learning by Teaching**
- **Justification:**
  The GridPane layout in JavaFX is essential for aligning UI components in a structured, row-column format. To help students grasp this layout more effectively, a Learning by Teaching activity was chosen. This strategy encouraged students to take ownership of the topic, prepare instructional material, and deliver it to classmates enhancing their learning experience and developing key communication competencies.

- **Time Allotted for the Activity:** 30 Minutes

- **Details of the Implementation:**
  - Learning by Teaching is a student-centered instructional method where learners develop a deeper understanding of a topic by preparing and teaching it to their peers.
  - For this activity, the topic JavaFX GridPane Layout was assigned to two students Ms. R. Rashika and Ms.J.S Divya Sri one week in advance to allow adequate preparation.
  - The Student Ms. R.Rashika began the session by explaining the development steps of a JavaFX GridPane Layout using a real time example on simple login form.
  - Following this, Ms.J.S Divya Sri wrote the complete JavaFX GridPane Layout program for simple login form on the board and explained it to peers shown in Figure1.
  - The entire session lasted approximately 30 minutes, covering the topic explanation, code demonstration, and a doubt-clearing session.

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO9	PO10	PSO1
CO5	3	2	3	2	2	3

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO9	PO10	PSO1
Justification for correlation	Students will be able to understand the concept of JavaFX GridPane layouts and apply it in UI design for real time applications	Students will be able to develop interactive GUI applications using JavaFX GridPane layout	Students will learn to select and use layout managers appropriately for real world scenarios.	Students will be able to improve individual communication by presenting the topic independently.	Students presentation skill will be improved as they are facing the audience and teaching the concepts	Students will be to apply GridPane in developing interactive software applications using JavaFX.

- **Images / Screenshot of the practice:**

Fig 1: Learning by Teaching Activity by R Rashika R and J.S Divya Sree

- **Reflective Critique:**

❖ ***Feedback of practice from students and other stakeholders:***

  - Students were attentive and expressed that the session helped them understand how GridPane arranges components in rows and columns.
  - The classroom environment was interactive, and students felt free to ask questions during the peer-teaching session.

❖ ***Benefit of the practice:***

- This activity strengthened technical knowledge and communication skills through peer interaction.
- It supported slow learners, allowing them to understand the topic from a peer perspective.
- It encouraged knowledge-sharing, fostering collaborative learning.

❖ ***Challenges faced in implementation:***

- Since the activity involved only two presenters, not all students were engaged in self-learning.
- Additional demonstration by the instructor was needed to ensure all students understood the GridPane layout clearly.
- The activity required additional time to allow students to share their views more effectively

The instructor concluded the session by reviewing the JavaFX GridPane layout and demonstrating its implementation with an example.

References:

- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/1_VA_CS8603_Learning_teaching.pdf
- https://www.ritrjpm.ac.in/images/computer-science/2022-2023/6_GS_CS8592_learning_by_teaching.pdf

Signature of Faculty Member **HOD**